gpu/nist: best of $P + M$ and $P - M$ (`RHO_BEST2`)

Study `research/nist_best2_small_batch_20261027`, branch `research/nist-best2-small-batch-20261027`. Iteration run `iterations-v1` on Modal CPU containers (16 vCPU, shared), 2026-10-27; GPU A/B `gpu-ab-v1` on one RTX PRO 6000 on Modal, 2026-10-27 (commit 52c83d5a0).

The kernel change computes both images of an ADD step from the one shared inverse (one extra multiplication and squaring) and keeps the one whose x limb 1 scores higher. CPU suites pass with and without it, and the toy-curve end-to-end solve recovers the planted logarithm through the new subtract path.

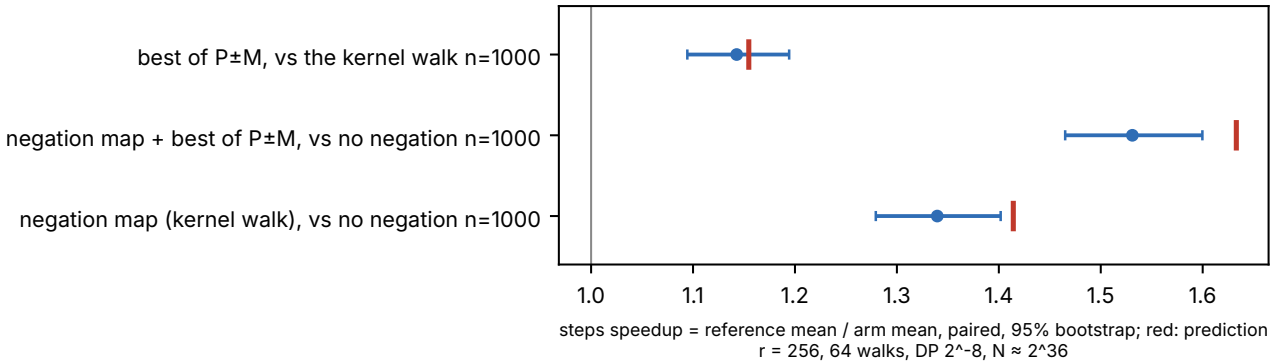


Figure 1: Steps to a useful collision, paired by trial (1,000 per arm), 95% bootstrap; red ticks are the derived predictions ($\sqrt{2}$ for the negation map, $\sqrt{4/3}$ for the selection). Rows against "no negation" sit below their prediction by the distinguished-point overhead (64 walks × 256). Source: `iterations-v1/summary*.jsonl`.

Measured: best of $P \pm M$ needs **1.143**× fewer steps than the kernel's negation-map walk (95% CI 1.094–1.194; predicted 1.155). Fruitless 2-cycle escapes rise from 1,931 to 2,570 per 10^6 steps.

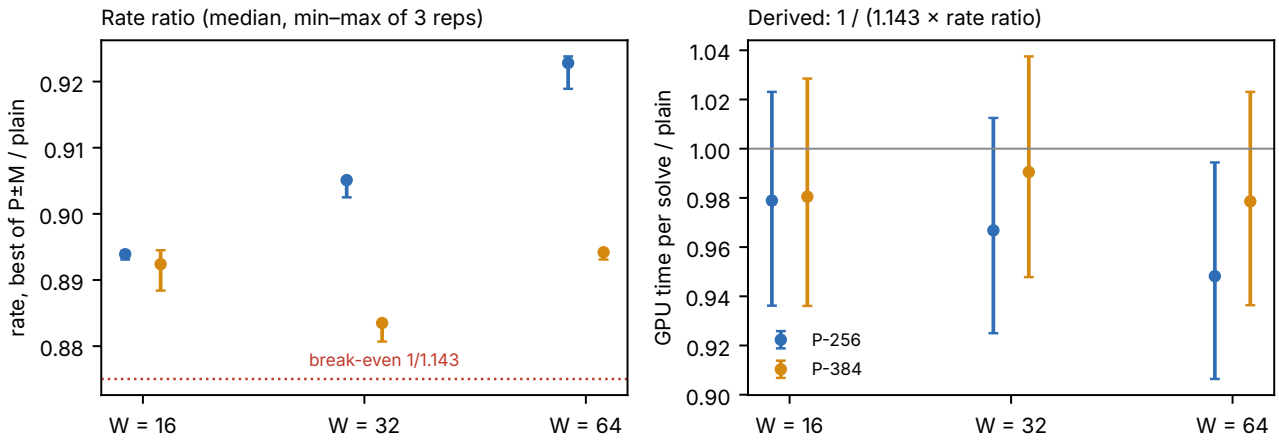


Figure 2: RTX PRO 6000 A/B (`gpu-ab-v1.summary.jsonl`): left, rate of best of $P \pm M$ over the plain kernel (median and min–max of 3 alternating repetitions; dotted: break-even $1/1.143$); right, derived GPU time per solve $1/(1.143 \times \text{rate ratio})$, the interval combining the iteration CI with the rate min–max.

Verdict. The selection costs 8–12% of the rate. Net GPU time per solve is 0.948–0.991 of the plain kernel. The clearest saving is P-256 at W = 64, the kernel's fastest cell: 0.948 (0.906–0.994). Other cells save 1–3% with intervals crossing 1. This is an engineering gain, not a change of exponent.

Open observations.

- With the selection, P-256 distinguished points ran 2–8% below the 2^{-24} expectation (0.923, 0.946, 0.974 at W = 16/32/64, about 2–3σ). P-384 and the plain build sit at the expectation. No cause is identified.
- The kernel's cycle-seek iterations add about 0.4% more GPU time for the selection (derived).
- The result stays in this research directory: the validation gate admits a `benchmarks/README.md` row only with a replay certificate and a validator report, and this run has neither yet.